

Troubleshooting WLAN Issues

AirWave Help Desk Guide

In a typical IT organization, it is the Help Desk's job to take incoming user support calls and determine whether the problem is an individual client/device issue or a broader network issue that might affect multiple users. The Help Desk itself is usually responsible for handling the individual user problems, while escalating broader network issues to the Network Engineering or Network Operations team.

With wireless networks, most user complaints boil down to one of two observable problems:

- The wireless network is slow.
- The wireless network is slow.

Of course, there are literally hundreds of different potential root causes for either of these two symptoms. Many, if not most, of these problems are related to the client device settings or authentication issues, which should be handled by the Help Desk. Yet, when the Help Desk does not have the tools and diagnostic capabilities to perform this 'triage,' most issues are instead escalated directly to Network Engineering. The result is not pretty: users are unhappy because their problems are not resolved quickly; the Help Desk staff becomes frustrated because they cannot do their jobs; and Network Engineers suffer because they are swamped with wireless related calls.

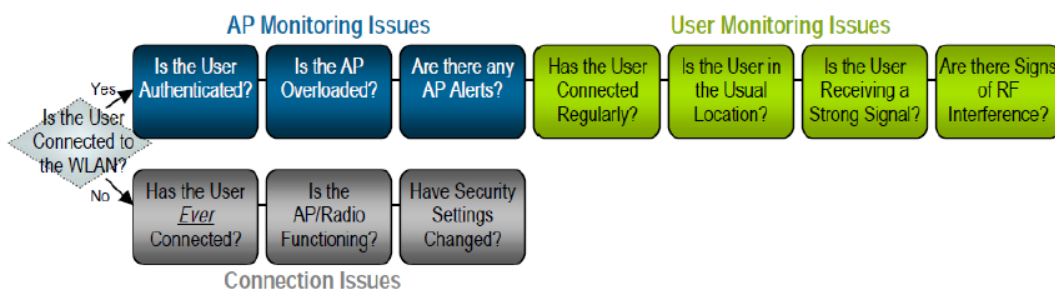
The AirWave Wireless Management Suite™ (AWMS) provides an end-to-end wireless operations management solution for the entire IT staff, from Network Engineering to the Help Desk. This quick reference guide is designed specifically to enable the Help Desk staff to:

Distinguish between common user/device problems and network issues:

- Diagnose and resolve client issues
- Diagnose and resolve client issues

Most of these steps can be performed in minutes with read-only access to the AirWave Management Platform™ (AMP) software, usually while the end user is still on the telephone. The basic troubleshooting workflow process for the Help Desk is depicted in the following image:

Figure 1 *Basic troubleshooting workflow*



Basic troubleshooting can be summarized in four steps:

- [Step One: Determine Whether the User is Connected](#)
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Step One: Determine Whether the User is Connected

The first step in the troubleshooting process is to determine whether the user is actually connected to the wireless network.

1. **Search by Username:** Ask the user for a user name, and enter it into AirWave's Search box.
2. **Determine the user's connection status:** Verify whether the user is currently connected to the WLAN. If the user is currently connected, the user name and session information will be shaded in green on AirWave's Search Results page. If the user is not connected, the information is italicized.
 - If the user is currently connected, click on the MAC Address link to open the diagnostic page for that client. Proceed to [Step Two: Check for Network-Related Issues](#).
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Figure 2 *Standard font indicates that the user is connected; italics indicate the user is not connected*

NAME	MAC ADDRESS	IP ADDRESS	DEVICE TYPE	AP	CONNECT TIME	MODE	FLOOR PL
<i>laura@hp.com</i>	<i>B4:86:76:52:80:E6</i>	-	<i>Windows 8</i>	<i>1344-1-AP13</i>	<i>0 seconds</i>	<i>11n 5 GHz (40)</i>	-
laura@hp.com	60:57:18:C0:8A:E6	15.110.208.141	Windows 7	1341-AP104	59 minutes	11ac 5GHz (80)	Sunnyval
laura@hp.com	BC:54:36:74:18:AF	FE80::100F:607A:1A28:3FB7	iPhone	1341-AP104	1 hour 0 minu...	11ac 5GHz (80)	Sunnyval

If the user has connected recently: Verify verbally that the user is in his or her usual location. Proceed to [Step Two: Check for Network-Related Issues](#). If you do not have a dense AP environment with overlapping coverage areas, a down AP can be the source of many end user trouble tickets.

If the user has not connected recently: Determine whether the user has changed hardware recently or whether your organization has changed security policies, passwords, etc. since his or her last connection. Many organizations are migrating from WEP to WPA or WPA2. Intermittent or infrequent wireless network users may not be aware of changes to security policies that can affect their ability to connect to the network.



If AirWave does not show that the user is connected to your wireless network, but the user reports that he/she has network access, the user could be connected to a rogue access point or unauthorized ad hoc network. In this case, the Help Desk should instruct the user to contact Network Engineering if connected to the right SSID.

Step Two: Check for Network-Related Issues

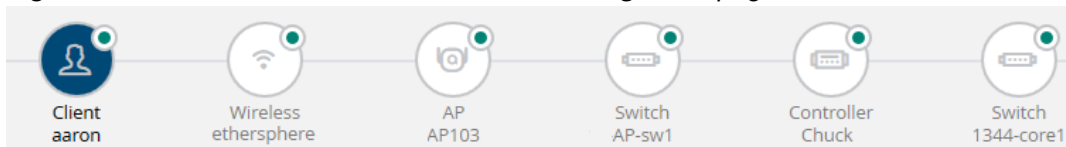
1. **Verify that the user is connected to the AP:** After verifying that the user is connected to the wireless network, verify that the user is connected to the access point. Click on the user's MAC

address. This takes you to the **Clients > Diagnostic** page for the client. If the user is connected, then this page includes the current connection table, as shown in the following table.



If the user is no longer connected, then selecting the MAC address link will take you to the Device Info page for the selected client rather than to the connection table.

Figure 3 Network information on the **Clients > Diagnostic** page



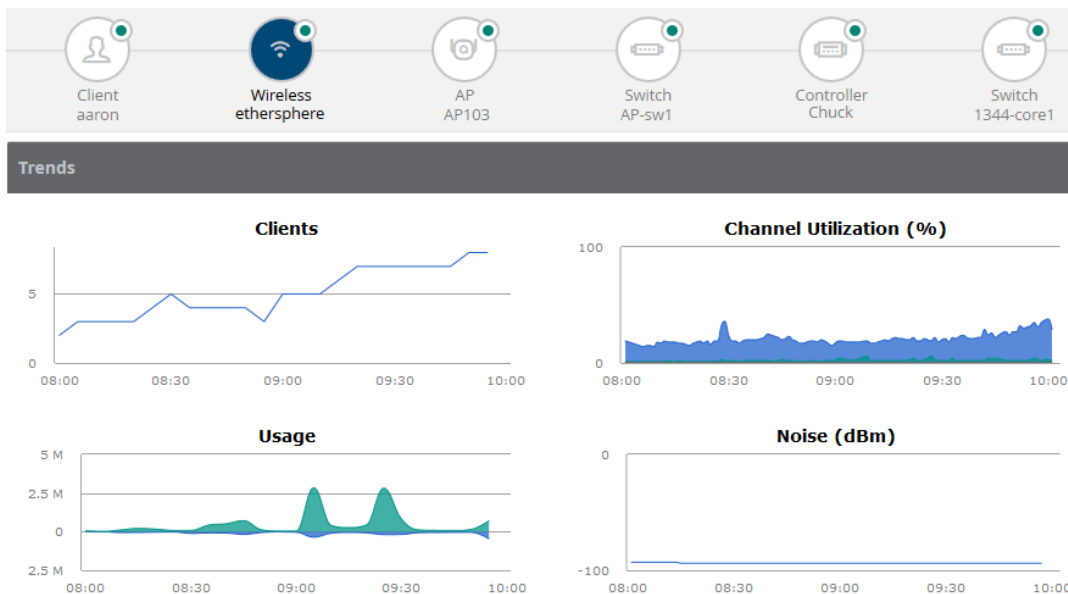
2. **Verify that the user is authenticated:** Review the information in the **Current Association** table on this page. Determine whether the user is currently authenticated by checking the Auth Type field, and by verifying whether the user has been assigned an IP address. If your organization is using multiple VLANs and SSIDs, the Help Desk should also be able to verify that the user is connected to the appropriate VLAN (i.e., an employee is not connected to a Guest VLAN). If the user does not appear to be authenticated, you should determine whether the user has the appropriate credentials that are required to connect to your WLAN.

Figure 4 Authentication information on the **Clients > Diagnostic** page

Current Association	
Connection Mode:	11n 5 GHz (40)
Duration:	4 hours 7 minutes
IP Address:	2620:11D:6010:B08:B...
Hostname:	
Usage:	3.39 Kbps
Security Mode:	WPA2
Cipher:	AES
Auth type:	EAP

3. **Check AP Usage Levels:** Check the current usage levels on the AP or radio to which the user is connected to determine whether the AP is overcrowded, resulting in poor performance. Select the AP in the current connection table to review the AP's Client and Usage information.

Figure 5 AP Client and Usage Data on the **Clients > Diagnostic** page



- If no other users are connected to that AP, then this can be a sign that something is wrong with the AP. Click in the Clients graph to open the graph popup. Select a time range above the graph to view historical information (2 hours, 1 day, 5 days, 1 month, or 1 year) and determine whether it is unusual for no other users to be connected at this time of day. If no other users are connected at a time when usage is normally high, it is more likely that there is an AP or radio problem that should be escalated to Network Engineering.
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If usage appears unusually high, you can suggest that the user reporting the problem connect to the wired network temporarily while usage is high. You might also elect to contact those users with the highest bandwidth utilization levels to determine whether they can shift to the wired network temporarily to relieve over-utilization of the wireless network. If the AP appears to be overloaded on a consistent basis when you look at the historical usage graphs, you might want to alert Network Engineering that there may be a need to add capacity, change RF transmission power, shift more users from 802.11 b/g to 802.11a, etc.

4. **Review the Indicators:** In the **Quality** table on this page, review the Indicators for the AP to determine if any issues may result in slow network connectivity. Each indicator includes a link that you can select to view additional, detailed information.

Figure 6 AP Indicators on the **Clients > Diagnostic** page

Quality		
Overall rating 		
ADDITIONAL INDICATORS (WIRELESS)		
INDICATOR	VALUE	IDEAL
Channel Utilization (%)	31.1%	≤ 60%
Noise Floor	-94 dBm	≤ -90 dBm
Avg. Frame Errors/Sec	81 frames/s	≤ 250 frames/s
Down Neighbors	0	0
Thresholds		

Step Three: Examining User Information

1. **Review the Matching Events:** In the **Clients > Diagnostic** page for the client, review the **Match Events** table for the client and check for frequent roams between access points. In this example, a user has moved to different APs several times. If the user is mobile, this roaming pattern may simply reflect physical movements. If the user has been stationary, however, this may indicate that the user is ping-ponging back and forth between two APs that are both within RF range, and this ping-ponging can explain certain performance problems. In this case, the Help Desk can decide to assist the user in changing the client device settings by putting the settings back to the default. If this problem seems to be affecting multiple users over an extended period of time, Network Engineering might need to adjust load-balancing settings. Similarly, if a user was associated to an AP that is within RF range but farther away than their usual AP, the user may be receiving a poor signal as a result. You can then help the user disassociate from their current AP and re-associate to the closer AP with a stronger signal. The Help Desk might need to talk to the admin if the client match settings need to be adjusted to be less aggressive.

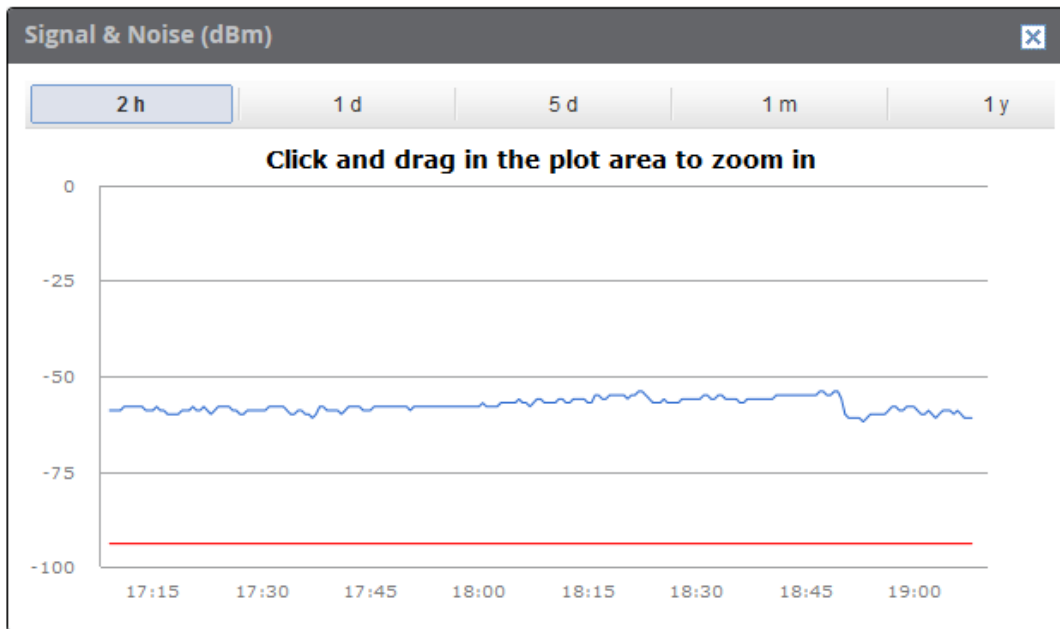
Figure 7 Match Events table on the **Clients > Diagnostics** page

Match Events			More
EVENT TIME ▼	FROM AP	TO AP	
1/19/16, 10:48 AM	AP103	1341-AP106	✓
1/19/16, 7:36 AM	AP106	1341-AP103	✓
1/19/16, 7:32 AM	AP103	1341-AP106	✓
1/19/16, 7:27 AM	AP106	1341-AP103	✓
1/19/16, 7:04 AM	AP103	1341-AP106	✓

2. **Check the user's signal quality:** A weak or poor quality RF signal may be the cause of unusually slow wireless network performance. Click in the **Signal & Noise** graph on the **Clients > Diagnostics** page to view the graph popup. Review the **Signal Quality** graph to determine whether the AP is receiving a strong RF signal from the user's client device. If the signal quality appears weak, and the user is connected to his or her usual AP, change the time range to compare the user's current signal quality to historical levels from the past day, 5 days, month, and year. In [Figure 8](#) below:

- The blue line is the Signal Level. Client devices on a correctly designed network should be -65 dBm or stronger (-64 and -63).
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Figure 8 *Signal Quality graphs for last 2 hours*



It is important to compare current RF signal quality levels to historical levels. If a user is complaining that the WLAN is unusually slow today, and the user's signal quality is low, there may be an RF problem affecting the user. However, if the user's signal quality is always low (for example, because his desk is 150' from the nearest wireless AP), then a low signal quality measure may not explain why the network performance today is worse than usual.

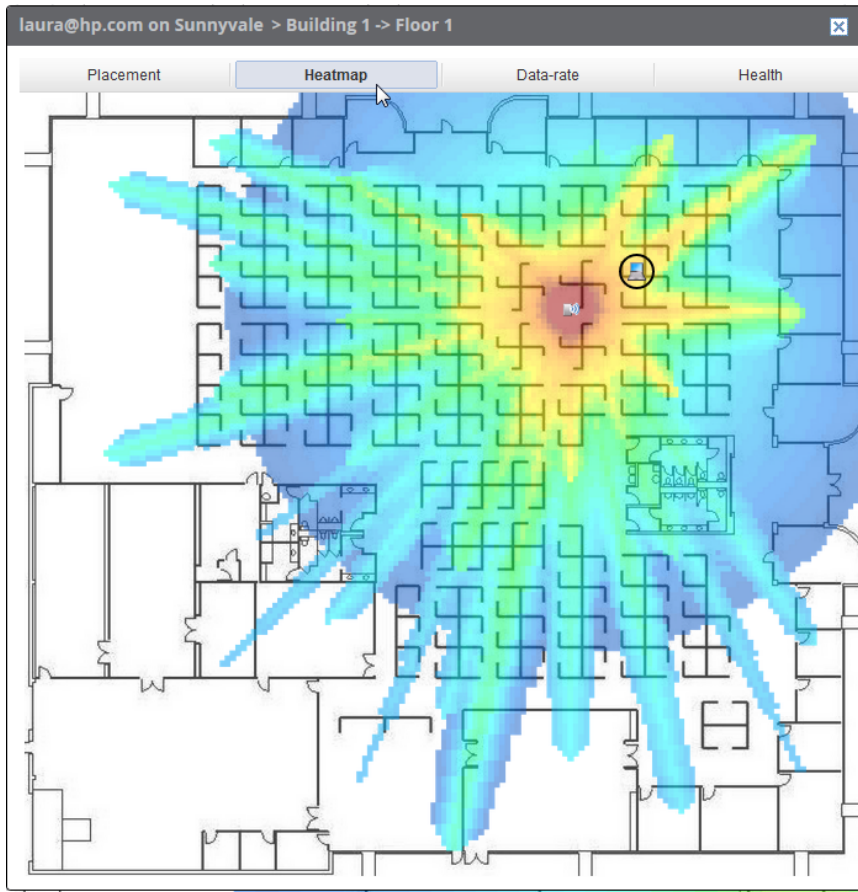
Step Four: Using Location Information, RF Heatmaps, and Frames/Sec

1. **Test Distance as a Cause of Poor Signal Quality:** If you suspect that poor RF signal quality might be the source of the user's problem, perform a simple test while the user is still on the phone to determine whether distance from a wireless AP is a major contributing factor.
 - a. Go to the **Clients > Diagnostics** page for the client. You can access this page after performing a search for a user, and selecting the associated MAC address from the search results.
 - b. Double-click the Floor Plan on the **Clients > Diagnostics** page to open the floor plan in a popup window.
 - c. Click the **Heatmap** tab.
 - d. Determine the physical location of the AP to which the user is connected.
 - e. If the user is not near the AP, ask the user to move closer to the AP and report whether performance improves.
 - If performance improves noticeably when the user is closer to the AP, then poor RF signal strength is a likely cause of the problem. The heatmap view in VisualRF can show you if any neighboring access points might provide a stronger, clearer signal to the user. In the example above, the areas with the strongest signal are depicted in red,

while areas with the weakest (or no) signal are in light blue (or white).

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Figure 9 User location information



2. **Review the Number of Frame Errors per Second:** A high error rate for Frames per Second can indicate that wireless traffic is not flowing smoothly.

- In the **Client > Diagnostics** page, click the network icon at the top of the page to display information about the network.
- In the **Quality** table click the **Indicat Avg. Frame Errors/Sec.** links. The **Avg Frame Errors/Sec Diagnostics** window opens.
- Check the number of FCS (Frame Check Sequence) Errors. If the error rate has been unusually high, it means that many wireless packets are being garbled. This is a clear indication of interference and a low signal quality. If error rates are high, this is important information to convey when escalating to Network Engineering. Keep in mind that it is important to compare current error levels to historical levels to see if current levels truly appear anomalous.

Figure 10 *Review the frames/sec information*

Avg. Frame Errors/Sec Diagnostics	
Indicator	Avg. Frame Errors/Sec
Value	49 frames/s
Ideal	≤250 frames/s
Explanation	MAC and PHY frame errors per second. High frame error rates indicate network problems.
Resolution	Check for interference or capacity issues.

Terminology Change

As part of advancing HPE's commitment to racial justice, we are taking a much-needed step in overhauling HPE engineering terminology to reflect our belief system of diversity and inclusion. Some legacy products and publications may continue to include terminology that seemingly evokes bias against specific groups of people. Such content is not representative of our HPE culture and moving forward, Aruba will replace racially insensitive terms and instead use the following new language:

Usage	Old Language	New Language
Campus Access Points + Controllers	Master-Slave	Conductor-Member
Instant Access Points	Master-Slave	Conductor-Member
Switch Stack	Master-Slave	Conductor-Member
Wireless LAN Controller	Mobility Master	Mobility Conductor
Firewall Configuration	Blacklist, Whitelist	Denylist, Allowlist
Types of Hackers	Black Hat, White Hat	Unethical, Ethical

Contacting Support

Main Site	arubanetworks.com
Support Site	asp.arubanetworks.com
Airheads Social Forums and Knowledge Base	community.arubanetworks.com
North American Telephone	1-800-943-4526 (Toll Free) 1-408-754-1200
International Telephone	arubanetworks.com/support-services/contact-support/
Software Licensing Site	lms.arubanetworks.com
End-of-life Information	arubanetworks.com/support-services/end-of-life/
Security Incident Response Team (SIRT)	Site: arubanetworks.com/support-services/security-bulletins/ Email: aruba-sirt@hpe.com



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